Physics-Informed DMD (piDMD)

Aim: Learn structured linear models from data.

Recap

Given snapshot matrices

$$X = \begin{bmatrix} | & & | \\ x_1 & \cdots & x_m \\ | & & | \end{bmatrix} \quad \text{and} \quad Y = \begin{bmatrix} | & & | \\ y_1 & \cdots & y_m \\ | & & | \end{bmatrix},$$

$d \times m$ $\qquad$ $d \times m$

where $\qquad y_j = T x_j, \qquad j = 1, \dots, m,$

piDMD finds the "best-fit" model in $\mathcal{M} \subset \mathbb{R}^{d \times d}$,

$$(*) \qquad \hat{T}_{PI} = \underset{S \in \mathcal{M}}{\operatorname{argmin}} \|Y - SX\|_F .$$

and $\mathcal{M}$ is chosen based on key physical properties.

Shift-invariant		Toeplitz/Circulant
Causal (1D)		Triangular
Conservative	$\Longleftrightarrow$	Unitary
Time-periodic		Eigvals on unit circle

⇒ See demo04.m for numerical experiments comparing DMD & piDMD for advection.

Q: How do we solve (*) to construct the piDMD model from snapshot data?

⇒ Depends strongly on $\mathcal{M}$. We can often construct $\hat{T}_{PI}$ and its eigendecomposition explicitly from X and Y, just as we do for the standard DMD problem.

⇒ More generally, can use constrained optimization algorithms designed for manifolds.

⇒ We'll discuss two important cases, shift-invariant systems & unitary systems. See the piDMD paper for more.

<u>Shift-Invariant Systems</u>

Consider a discrete shift-invariant system:

$$x_{k+1} = T x_k, \qquad k = 0, 1, 2, \ldots$$

where T commutes w/ the periodic shift:

$$S = \begin{bmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & \ddots & \ddots \\ & & & 1 \\ 1 & & & 0 \end{bmatrix}, \qquad TS = ST.$$

The commutation property implies that T is circulant, i.e., $T = STS^*$ implies that T is constant along diagonals, with

$$T = \begin{bmatrix} \tau_0 & \tau_1 & \cdots & \tau_{d-1} \\ \tau_{d-1} & \tau_0 & \tau_1 & \vdots \\ \vdots & \ddots & \ddots & \ddots \\ & & & \tau_1 \\ \tau_1 & \cdots & \tau_{d-1} & \tau_0 \end{bmatrix}.$$

Consequently, we can write T as a linear combination of powers of the shift operator:

$$T = t_0 I + t_1 S + \cdots + t_{d-1} S^{d-1}.$$

By the spectral mapping theorem, the eigenvectors of T are the eigenvectors of S, and the eigenvalues of T are related to those of S:

$$\lambda_j(T) = t_0 \lambda_j(S) + t_1 \lambda_j(S)^2 + \cdots + t_{d-1} \lambda_j(S)^{d-1}$$

for $j = 1, 2, \ldots, d$. The eigenvectors of S are simply columns of the Discrete Fourier Transform (DFT) matrix F, since

$$S \begin{bmatrix} 1 \\ \omega^j \\ \vdots \\ \omega^{(d-1)j} \end{bmatrix} = \begin{bmatrix} \omega^j \\ \vdots \\ \omega^{(d-1)j} \\ 1 \end{bmatrix} = \omega^j \begin{bmatrix} 1 \\ \omega^j \\ \vdots \\ \omega^{(d-1)j} \end{bmatrix}, \quad \omega = e^{-2\pi i/d}$$

Normalizing each eigenvector, we have

$$S = F \underset{\substack{\| \\ \operatorname{diag}(1, \omega, \ldots, \omega^{d-1})}}{\Omega} F^*, \quad \text{where} \quad F = \frac{1}{\sqrt{d}} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & \omega & \cdots & \omega^d \\ \vdots & \vdots & & \\ 1 & \omega^d & \cdots & \omega^{d^2} \end{bmatrix}.$$

Therefore, the eigendecomposition of T is

$$T = F(t_0 \Omega + \cdots + t_{d-1} \Omega^{d-1}) F^*.$$

Q: How do we construct the "best" circulant matrix to fit the snapshot data?

(*) $$T = \underset{T = \text{circulant}}{\arg\min} \|Y - TX\|_F$$ PI

We can use the fact that every circulant matrix is diagonalized by the DFT to decouple the objective function in (*).

Since F is unitary, we have that

$$\|Y - TX\|_F = \|F^*Y - F^*TX\|_F = \|F^*Y - \Lambda F^*X\|_F,$$

where $T = F\Lambda F^*$ is the eigendecomp of T. Define $\tilde{Y} = F^*Y$ and $\tilde{X} = F^*Y$ to be the Fourier transforms of the snapshot matrices,

$$\|\tilde{Y} - \Lambda\tilde{X}\|_F^2 = \sum_{i=1}^{d}\sum_{j=1}^{m}(\tilde{y}_{ij} - \lambda_i\tilde{X}_{ij})^2 = \sum_{i=1}^{d}\|\tilde{\underline{y}}_i - \lambda_i\tilde{\underline{x}}_i\|_2^2,$$

where $\tilde{\underline{y}}_i$ and $\tilde{\underline{x}}_i$ are the i^{th} rows of $\tilde{Y}$ and $\tilde{X}$, respectively. Note that the i^{th} term in the sum depends only on λ_i, so we can choose each λ_i to minimize the i^{th} term indep. of the other $d-1$ terms. The i^{th} term is just an overdetermined least-squares problem, so we can apply the orthogonality criterion and solve for λ_i:

$$\lambda_i = \frac{\tilde{y}_i \tilde{x}_i^*}{\tilde{x}_i \tilde{x}_i^*} = \frac{\tilde{y}_i \tilde{x}_i^*}{\|\tilde{x}_i\|^2}, \qquad i=1,\dots,d.$$

This determines the eigenvalues of the "best" circulant matrix for the snapshots:

$$T = F\Lambda F^*, \quad \Lambda = \mathrm{diag}(\lambda_1,\dots,\lambda_d).$$

In summary, the piDMD algorithm for shift-invariant systems is:

<u>Step 1.</u> Compute $\tilde{X} = F^* X$, $\tilde{Y} = F^* Y$.

<u>Step 2.</u> Compute $\lambda_i = \tilde{y}_i \tilde{x}_i^* \|\tilde{x}_i\|^{-1}$, $i=1,\dots,d$
where $\tilde{x}_i$ = row i of $\tilde{X}$, $\tilde{y}_i$ = row i of $\tilde{Y}$.

<u>Step 3.</u> Return $T = F\Lambda F^*$.

Note: In practice, may want to avoid forming T or F explicitly, since F and F^* have fast $O(d\log d)$ mat-vecs via the Fast Fourier Transform (FFT).

Conservative Systems

Consider a discrete conservative system:

$$x_{k+1} = T x_k, \qquad k=0,1,2,\dots$$

where $\|x_{k+1}\|_2 = \|x_k\|_2$ for all $k=0,1,2,\dots$
This implies that T is unitary, i.e, that

$$T^*T = TT^* = I.$$

We are looking for the unitary matrix T_{PI} that "best" fits the snapshot data:

$$(*) \qquad T_{PI} = \underset{T=\text{Unitary}}{\operatorname{argmin}} \|Y - TX\|_F.$$

To minimize $\|Y - TX\|_F$, expand

$$\|Y-TX\|_F^2 = \mathrm{tr}\left[(Y-TX)^T(Y-TX)\right] \quad \left(\begin{matrix}\text{c.f. lecture}\\ \text{11 notes}\end{matrix}\right)$$

$$= \mathrm{tr}\left[Y^TY\right] - 2\mathrm{tr}\left[Y^TTX\right] + \mathrm{tr}\left[X^TT^TTX\right]$$

Since $\mathrm{tr}\left[Y^TY\right] = \text{const.}$ and $\mathrm{tr}\left[X^TT^TTX\right] = \mathrm{tr}\left[T^TTX^TX\right] = \mathrm{tr}\left[T^TT\right] = \text{const.}$ (by the cyclical property of the trace), minimizing $\|Y-TF\|_F$ is equivalent to maximizing:

$$T_{PI} = \underset{T=\text{unitary}}{\mathrm{argmin}} \|Y-TX\|_F = \underset{T=\text{unitary}}{\mathrm{argmax}}\ 2\mathrm{tr}\left[Y^TTX\right].$$

We use cyclicity of the trace again,

$$\mathrm{tr}\left[Y^TTX\right] = \mathrm{tr}\left[TXY^T\right],$$

and substitute the SVD of $XY^T = U\Sigma V^T$,

$$\mathrm{tr}\left[TXY^T\right] = \mathrm{tr}\left[TU\Sigma V^T\right] = \mathrm{tr}\left[V^TTU\Sigma\right].$$

Since, U, V, and T are orthogonal, the matrix $Z = V^T T U$ is also orthogonal. In particular, its columns have length less than 1, so (since $\sigma_i = \Sigma_{ii} \geq 0$)

$$\mathrm{tr}\left[Z\Sigma\right] = \sum_{i=1}^{d} z_{ii}\sigma_i \leq \sum_{i=1}^{d} \sigma_i .$$

The maximum (equality) is achieved when $z_{ii} = 1$ for $i = 1, \ldots, d$, in which case $Z = I$, so $T = VU^T$. Therefore, the best unitary/orthogonal fit to the data is $T_{PI} = VU^T$, where $XY^T = U\Sigma V^T$.

Note: The matrix $T_{PI} = VU^T$ is the transpose of the polar factor in the polar decomposition of the matrix XY^T, $XY^T = \underbrace{(UV^T)}_{\text{unitary}}\underbrace{(V\Sigma V^T)}_{\text{SPD}}$

Step 1. Compute $XY^T = U\Sigma V^T$

Step 2. Compute $T_{PI} = VU^T$

Q: Can you combine the piDMD algorithms for shift-invariant & conservative systems to compute the best unitary circulant fit to the snapshot matrices?

Hint: See demo04.m